

|| Jai Sri Gurudev ||
Sri Adichunchanagiri Shikshana Trust (R)

SJB INSTITUTE OF TECHNOLOGY

Affiliated to Visvesvaraya Technological University, Belagavi & Approved by AICTE,
New Delhi Accredited by NAAC, New Delhi with 'A' Grade, Recognized by UGC, New
Delhi with 2(f) and 12(B) Certified by ISO 9001- 2015

BGSHealth & Education City, Dr. Vishnuvardhan Road, Kengeri, Bengaluru – 560 060



Study Material

Subject Name: COMPUTER NETWORKS (CN)

Subject Code: 21CS52

By

Faculty Name: Prof Chaithra A S & Prof. Kusuma C

Designation: Assistant Professor

Semester: 5th 'A' & 'B'



Department of Information Science & Engineering

Aca. Year: ODD Semester/2023-24

MODULE 1

Module 1

INTRODUCTION TO NETWORKS

1.1 INTRODUCTION

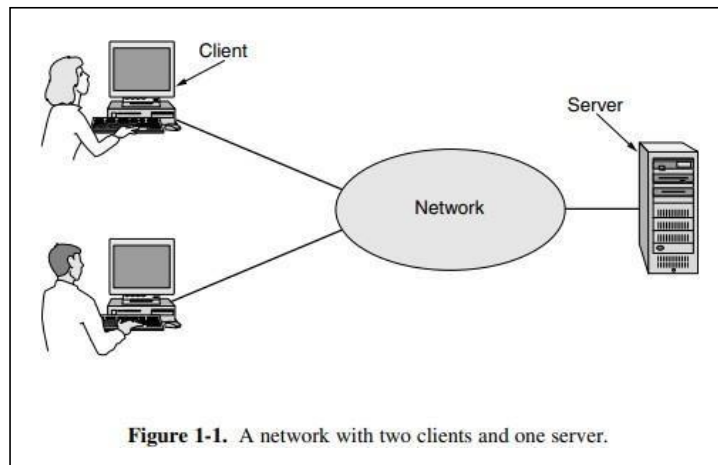
Computers have made spectacular progress in a short time. During the first two decades of their existence, computer systems were highly centralized, usually within a single large room. A medium-sized company or university might have had one or two computers, while very large institutions had at most a few dozen. The idea that within forty years vastly more powerful computers smaller than postage stamps would be mass produced by the billions was pure science fiction.

The merging of computers and communications has had a profound influence on the way computer systems are organized. The once-dominant concept of the “computer center” as a room with a large computer to which users bring their work for processing is now totally obsolete. The old model of a single computer serving all of the organization’s computational needs has been replaced by one in which a large number of separate but interconnected computers do the job. These systems are called **computer networks**.

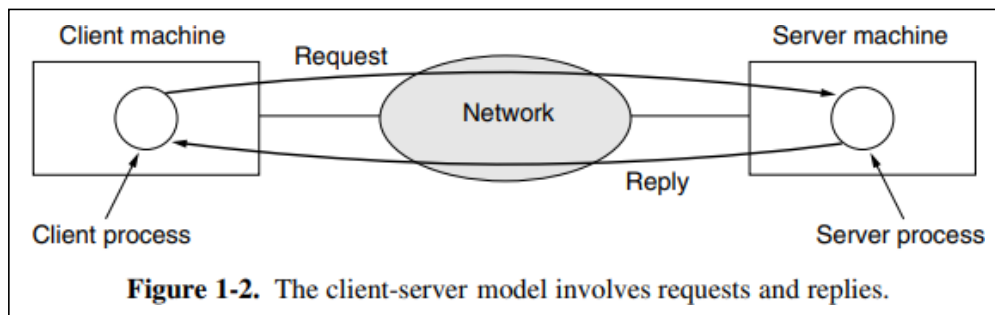
“**Computer network**” means a collection of autonomous computers interconnected by a single technology. Two computers are said to be interconnected if they are able to exchange information. The connection need not be via a copper wire; fiber optics, microwaves, infrared, and communication satellites can also be used. Networks come in many sizes, shapes and forms, as we will see later. They are usually connected together to make larger networks, with the Internet being the most well-known example of a network of networks.

Networks called **VPNs (Virtual Private Networks)** may be used to join the individual networks at different sites into one extended network. In other words, the mere fact that a user happens to be 15,000 km away from his data should not prevent him from using the data as though they were local. In the simplest of terms, one can imagine a company’s information system as consisting of one or more databases with company information and some number of employees who need to access them remotely. In this model, the data are stored on powerful computers called **servers**. Often these are centrally housed and maintained by a **system administrator**. In contrast, the employees have simpler machines, called **clients**, on their

desks, with which they access remote data, The client and server machines are connected by a network, as illustrated in Fig. 1-1. This whole arrangement is called the **client-server model**.



If we look at the client-server model in detail, we see those two processes (i.e., running programs) are involved, one on the client machine and one on the server machine. Communication takes the form of the client process sending a message over the network to the server process. The client process then waits for a reply message. When the server process gets the request, it performs the requested work or looks up the requested data and sends back a reply. These messages are shown in Fig. 1-2



Telephone calls between employees may be carried by the computer network instead of by the phone company. This technology is called **IP telephony or Voice over IP (VoIP)** when Internet technology is used. The microphone and speaker at each end may belong to a VoIP-enabled phone or the employee's computer. Companies find this a wonderful way to save on their telephone bills

1.2 NETWORK HARDWARE

Broadly speaking, there are two types of transmission technology that are in widespread use: **broadcast links** and **point-to-point links**.

Point-to-point links connect individual pairs of machines. To go from the source to the destination on a network made up of point-to-point links, short messages, called **packets** may have to first visit one or more intermediate machines. Often multiple routes, of different lengths, are possible, so finding good ones is important in point-to-point networks. Point-to-point transmission with exactly one sender and exactly one receiver is sometimes called **unicasting**.

In contrast, on a broadcast network, the communication channel is shared by all the machines on the network; packets sent by any machine are received by all the others. An address field within each packet specifies the intended recipient. Upon receiving a packet, a machine checks the address field. If the packet is intended for the receiving machine, that machine processes the packet; if the packet is intended for some other machine, it is just ignored.

Broadcast systems usually also allow the possibility of addressing a packet to all destinations by using a special code in the address field. When a packet with this code is transmitted, it is received and processed by every machine on the network. This mode of operation is called **broadcasting**. Some broadcast systems also support transmission to a subset of the machines, which known as **multicasting**.

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

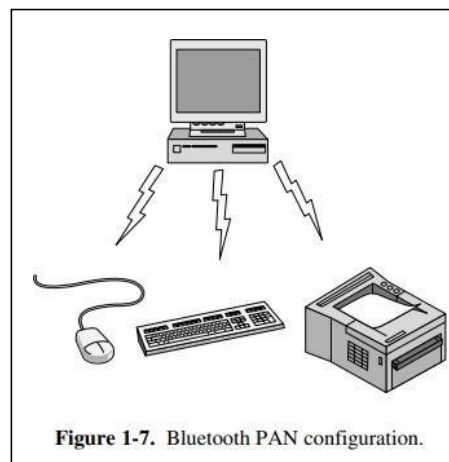
Figure 1-6. Classification of interconnected processors by scale.

An alternative criterion for classifying networks is by scale. Distance is important as a classification metric because different technologies are used at different scales. In Fig. 1-6 we classify multiple processor systems by their rough physical size. At the top are the **personal area networks**, networks that are meant for one person. Beyond these come longer-range networks. These can be divided into local, metropolitan, and wide area networks, each with increasing scale. Finally, the connection of two or more networks is called an **internetwork**.

The worldwide Internet is certainly the best-known (but not the only) example of an internetwork. Soon we will have even larger internetworks with the **Interplanetary Internet** that connects networks across space (Burleigh et al., 2003).

1.2.1 Personal Area Networks

PANs (Personal Area Networks) let devices communicate over the range of a person. A common example is a wireless network that connects a computer with its peripherals. Almost every computer has an attached monitor, keyboard, mouse, and printer. This connection must be done with cables. Some companies got together to design a short-range wireless network called **Bluetooth** to connect these components without wires. The idea is that if your devices have Bluetooth, then you need no cables. You just put them down, turn them on, and they work together.



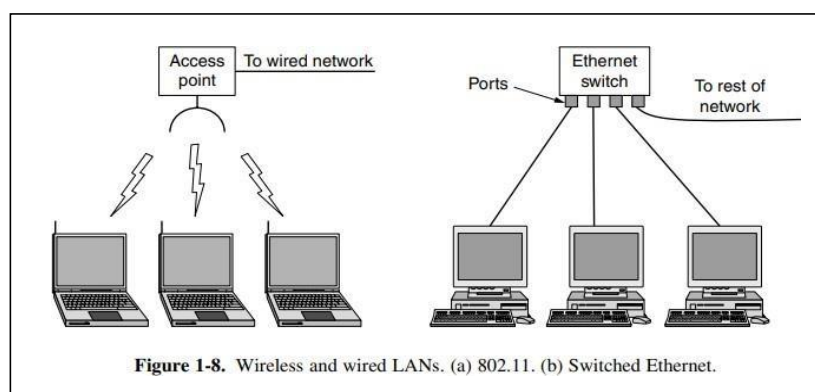
Bluetooth networks use the **master-slave** paradigm of Fig. 1-7. The system unit (the PC) is normally the master, talking to the mouse, keyboard, etc., as slaves. The master tells the slaves what addresses to use, when they can broadcast, how long they can transmit, what frequencies they can use, and so on.

PANs can also be built with other technologies that communicate over short ranges, such as **RFID** on smartcards and library books.

1.2.2 Local Area Networks

A **LAN (Local Area Network)** is a privately owned network that operates within and nearby a single building like a home, office or factory. LANs are widely used to connect personal computers and consumer electronics to let them share resources (e.g., printers) and exchange information. When LANs are used by companies, they are called **enterprise network**.

Wireless LANs are very popular these days, especially in homes, older office buildings, cafeterias, and other places where it is too much trouble to install cables. In these systems, every computer has a radio modem and an antenna that it uses to communicate with other computers. In most cases, each computer talks to a device in the ceiling as shown in Fig. 1-8(a). This device, called an **AP (Access Point)**, **wireless router**, or **base station**, relays packets between the wireless computers and also between them and the Internet.



There is a standard for wireless LANs called IEEE 802.11, popularly known as **WiFi**, which has become very widespread. It runs at speeds anywhere from 11 to hundreds of Mbps.

Wired LANs use a range of different transmission technologies. Most of them use copper wires, but some use optical fiber. LANs are restricted in size, which means that the worst-case transmission time is bounded and known in advance. Typically, wired LANs run at speeds of 100 Mbps to 1 Gbps, have low delay (microseconds or nanoseconds), and make very few errors. Newer LANs can operate at up to 10 Gbps. Compared to wireless networks, wired LANs exceed them in all dimensions of performance.

The topology of many wired LANs is built from point-to-point links. **IEEE 802.3**, popularly called **Ethernet**, is, by far, the most common type of wired LAN. Fig. 1-8(b) shows a sample topology of switched Ethernet. Each computer speaks the Ethernet protocol and connects to a box called a **switch** with a point-to-point link. Hence the name. A switch has multiple ports, each of which can connect to one computer. The job of the switch is to relay packets between computers that are attached to it, using the address in each packet to determine which computer to send it to. To build larger LANs, switches can be plugged into each other using their ports. Both wireless and wired broadcast networks can be divided into **static** and **dynamic designs**, depending on how the channel is allocated. A typical static allocation would be to divide time into discrete intervals and use a round-robin algorithm, allowing each machine to broadcast only when its time slot comes up. Static allocation wastes channel capacity when a machine

has nothing to say during its allocated slot, so most systems attempt to allocate the channel dynamically (i.e., on demand).

Dynamic allocation methods for a common channel are either **centralized** or **decentralized**. In the centralized channel allocation method, there is a single entity, for example, the base station in cellular networks, which determines who goes next. It might do this by accepting multiple packets and prioritizing them according to some internal algorithm. In the decentralized channel allocation method, there is no central entity; each machine must decide for itself whether to transmit.

1.2.3 Metropolitan Area Networks

A **MAN (Metropolitan Area Network)** covers a city. The best-known examples of MANs are the cable television networks available in many cities. A MAN might look something like the system shown in Fig. 1-9. In this figure we see both television signals and Internet being fed into the centralized cable headend for subsequent distribution to people's homes.

Recent developments in high-speed wireless Internet access have resulted in another MAN, which has been standardized as **IEEE 802.16** and is popularly known as **WiMAX**.

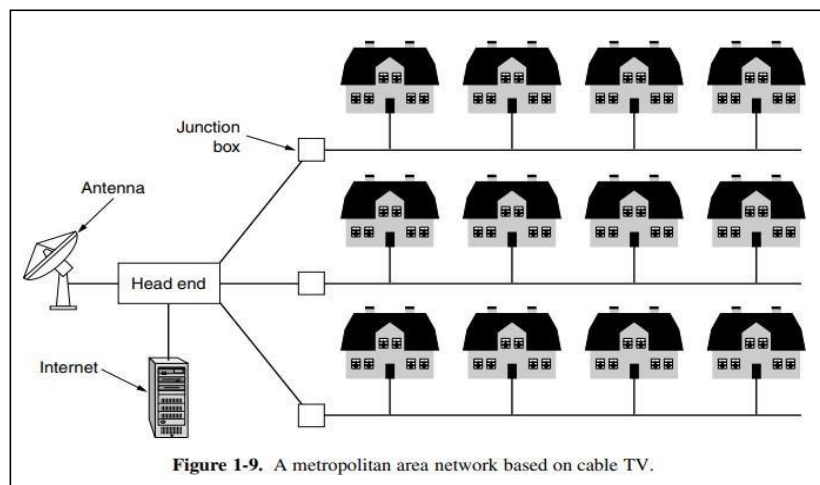
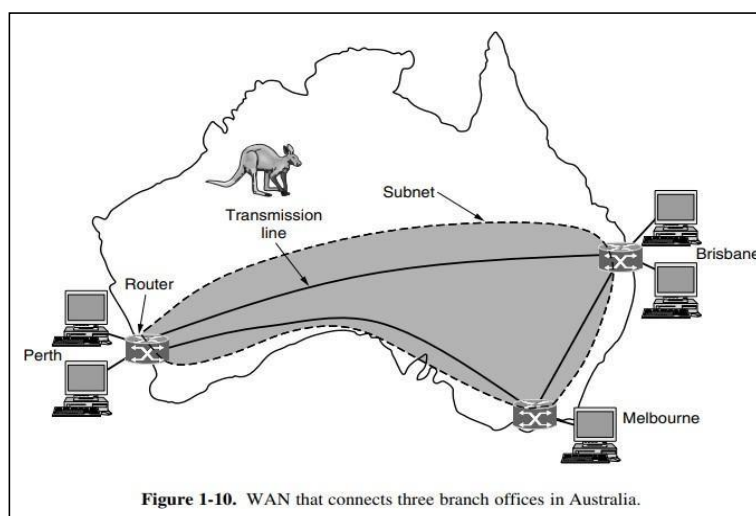


Figure 1-9. A metropolitan area network based on cable TV.

1.2.4 Wide Area Networks

A **WAN (Wide Area Network)** spans a large geographical area, often a country or continent. The WAN in Fig. 1-10 is a network that connects offices in Perth, Melbourne, and Brisbane. Each of these offices contains computers intended for running user (i.e., application) programs. These machines are called **hosts**. The rest of the network that connects these hosts is then called

the **communication subnet**, or just **subnet** for short. The job of the subnet is to carry messages from host to host.



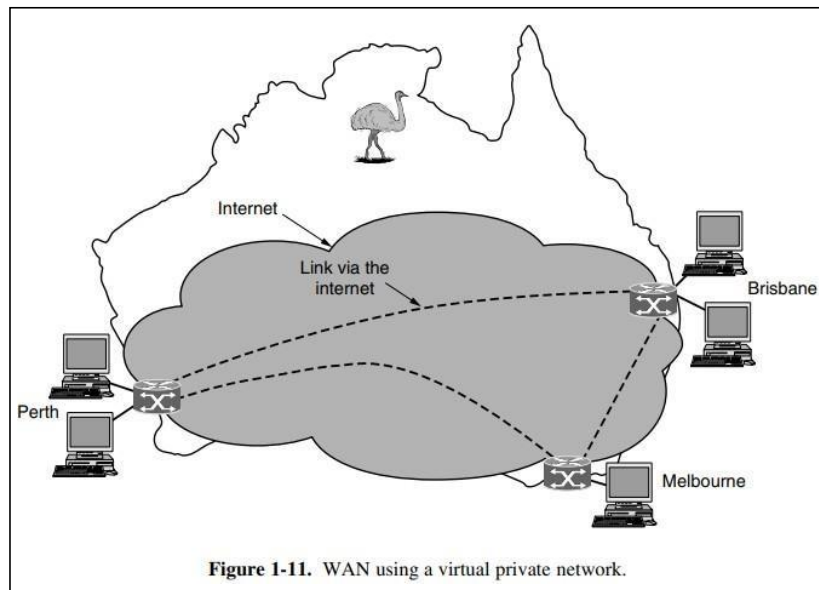
In most WANs, the subnet consists of two distinct components: transmission lines and switching elements. **Transmission lines** move bits between machines. They can be made of copper wire, optical fiber, or even radio links. **Switching elements**, or just switches, are specialized computers that connect two or more transmission lines. When data arrive on an incoming line, the switching element must choose an outgoing line on which to forward them. These switching computers are also called by the name **router**.

The WAN is similar to a large wired LAN, but there are some important differences. Usually in a WAN, the hosts and subnet are owned and operated by different people. A second difference is that the routers will usually connect different kinds of networking technology. The networks inside the offices may be switched Ethernet, for example, while the long-distance transmission lines may be SONET links. This means that many WANs will in fact be **internetworks**, or composite networks that are made up of more than one network. A final difference is in what is connected to the subnet. This could be individual computers, or it could be entire LANs.

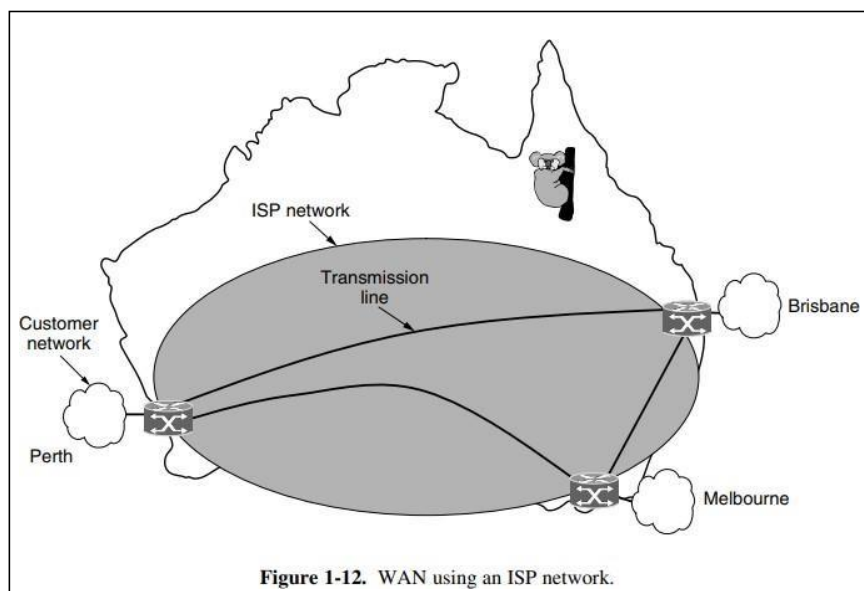
There are two other varieties of WANs. First, rather than lease dedicated transmission lines, a company might connect its offices to the Internet. This allows connections to be made between the offices as virtual links that use the underlying capacity of the Internet. This arrangement, shown in Fig. 1-11, is called a **VPN (Virtual Private Network)**.

A VPN has the usual advantage of virtualization, i.e; it provides flexible reuse of a resource (Internet connectivity). It is very easy to add a fourth office to see this. A VPN also has the usual disadvantage of virtualization, which is a lack of control over the underlying resources.

With a dedicated line, the capacity is clear. With a VPN your mileage may vary with your Internet service.



The second variation is that the subnet may be run by a different company. The subnet operator is known as a **network service provider** and the offices are its customers. This structure is shown in Fig. 1-12. The subnet operator will connect to other customers too, as long as they can pay and it can provide service. The subnet operator will also connect to other networks that are part of the Internet. Such a subnet operator is called an **ISP (Internet Service Provider)** and the subnet is an **ISP network**. Its customers who connect to the ISP receive Internet service.



In most WANs, the network contains many transmission lines, each connecting a pair of routers. If two routers that do not share a transmission line wish to communicate, they must do

this indirectly, via other routers. There may be many paths in the network that connect these two routers. The decision about which path to use is done using **routing algorithm**. Each router makes the decision about where to send packets next using **Forwarding algorithm**.

1.2.5 Internetworks

A collection of interconnected networks is called an **internetwork or internet**. The Internet uses ISP networks to connect enterprise networks, home networks, and many other networks. The term “**subnet**” refers to the collection of routers and communication lines owned by the network operator. As an analogy, the telephone system consists of telephone switching offices connected to one another by high-speed lines, and to houses and businesses by low-speed lines. These lines and equipment, owned and managed by the telephone company, form the subnet of the telephone system.

A **network** is formed by the combination of a subnet and its hosts. An internet is formed when distinct networks are interconnected. There are two rules of thumb that are useful in the case of Internetworks. First, if different organizations have paid to construct different parts of the network and each maintains its part, we have an internetwork rather than a single network. Second, if the underlying technology is different in different parts (e.g., broadcast versus point-to-point and wired versus wireless), we probably have an internetwork.

The general name for a machine that makes a connection between two or more networks and provides the necessary translation, both in terms of hardware and software, is a **gateway**. Gateways are distinguished by the layer at which they operate in the protocol hierarchy.

1.3 NETWORK SOFTWARE

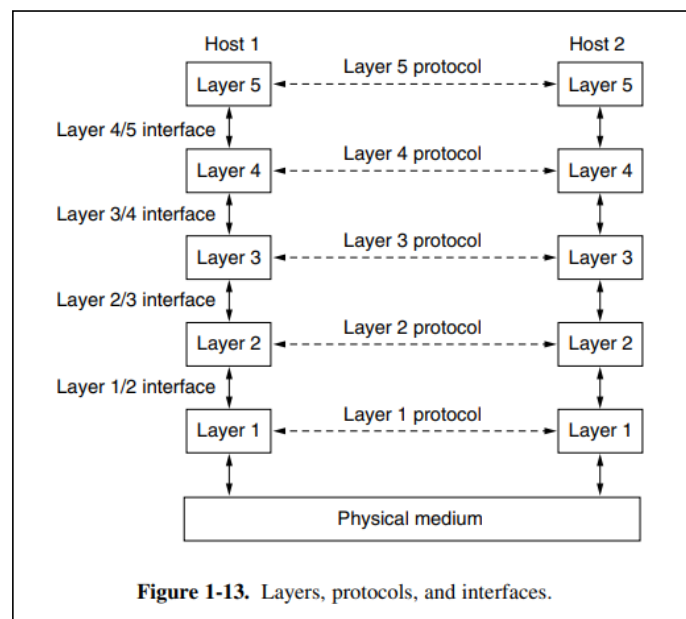
The first computer networks were designed with the hardware as the main concern and the software as an afterthought. This strategy no longer works. Network software is now highly structured. In the following sections we examine the software structuring technique in some detail.

1.3.1 Protocol Hierarchies

To reduce their design complexity, most networks are organized as a stack of **layers** or **levels**, each one built upon the one below it. The number of layers, the name of each layer, the contents of each layer, and the function of each layer differ from network to network. The purpose of each layer is to offer certain services to the higher layers while shielding those layers from the details of how the offered services are actually implemented. In a sense, each layer is a kind of virtual machine, offering certain services to the layer above it.

When layer n on one machine carries on a conversation with layer n on another machine, the rules and conventions used in this conversation are collectively known as the **layer n protocol**. Basically, a **protocol** is an agreement between the communicating parties on how communication is to proceed.

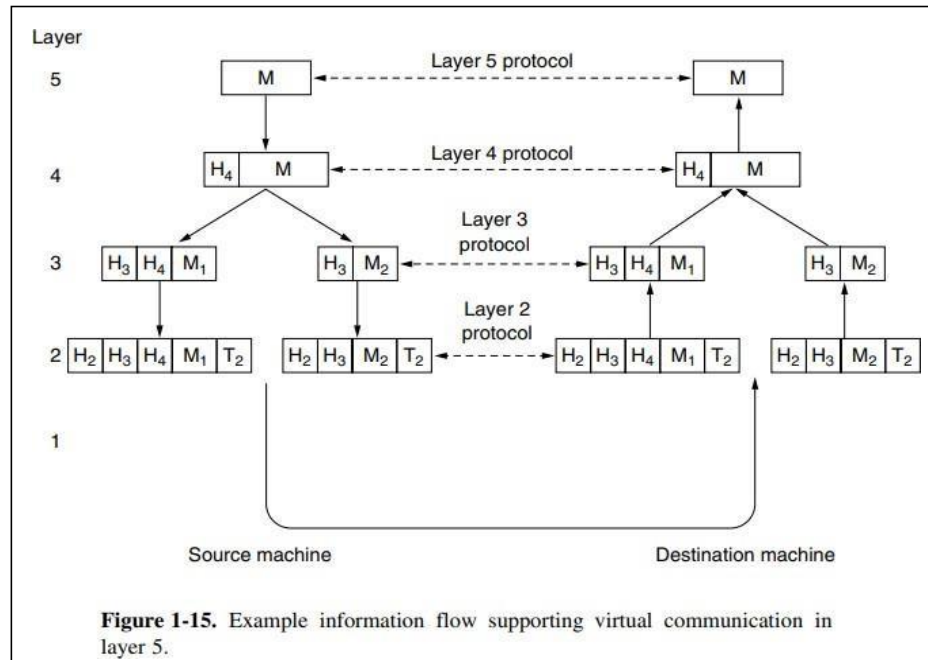
A five-layer network is illustrated in Fig. 1-13. The entities comprising the corresponding layers on different machines are called **peers**. The peers may be software processes, hardware devices, or even human beings. In other words, it is the peers that communicate by using the protocol to talk to each other.



In reality, no data are directly transferred from layer n on one machine to layer n on another machine. Instead, each layer passes data and control information to the layer immediately below it, until the lowest layer is reached. Below layer 1 is the physical medium through which actual communication occurs. In Fig. 1-13, virtual communication is shown by dotted lines and physical communication by solid lines.

Between each pair of adjacent layers is an **interface**. The interface defines which primitive operations and services the lower layer makes available to the upper one. When network designers decide how many layers to include in a network and what each one should do, one of the most important considerations is defining clean interfaces between the layers. In addition to minimizing the amount of information that must be passed between layers, clear cut interfaces also make it simpler to replace one layer with a completely different protocol or implementation (e.g., replacing all the telephone lines by satellite channels).

A set of layers and protocols is called a **network architecture**. The specification of an architecture must contain enough information to allow an implementer to write the program or build the hardware for each layer so that it will correctly obey the appropriate protocol. A list of the protocols used by a certain system, one protocol per layer, is called a **protocol stack**.



Consider a more technical example: how to provide communication to the top layer of the five-layer network in Fig. 1-15. A message, M , is produced by an application process running in layer 5 and given to layer 4 for transmission. Layer 4 puts a header in front of the message to identify the message and passes the result to layer 3. The header includes control information, such as addresses, to allow layer 4 on the destination machine to deliver the message.

In many networks, no limit is placed on the size of messages transmitted in the layer 4 protocol but there is nearly always a limit imposed by the layer 3 protocol. Consequently, layer 3 must break up the incoming messages into smaller units, packets, prepending a layer 3 header to each packet. In this example, M is split into two parts, M_1 and M_2 , that will be transmitted separately. Layer 3 decides which of the outgoing lines to use and passes the packets to layer 2. Layer 2 adds to each piece not only a header but also a trailer, and gives the resulting unit to layer 1 for physical transmission. At the receiving machine the message moves upward, from layer to layer, with headers being stripped off as it progresses. None of the headers for layers below n are passed up to layer n .

1.3.2 Design Issues for the Layers

One of the design issues is **Reliability**. Think about the bits of a packet traveling through the network. There is a chance that some of these bits will be received damaged (inverted) due to fluke electrical noise, random wireless signals, hardware flaws, software bugs and so on. We need to find and fix these errors.

One mechanism for finding errors in received information uses codes for **error detection**. Information that is incorrectly received can then be retransmitted until it is received correctly. More powerful codes allow for error correction, where the correct message is recovered from the possibly incorrect bits that were originally received. Both of these mechanisms work by adding redundant information. They are used at low layers, to protect packets sent over individual links, and high layers, to check that the right contents were received.

Another reliability issue is finding a working path through a network. Often there are multiple paths between a source and destination, and in a large network, there may be some links or routers that are broken. Depending upon the condition, the network should automatically make decision which alternate path needs to be taken to send the packets. This is called **routing**.

A second design issue concerns the evolution of the network. Over time, networks grow larger and new designs emerge that need to be connected to the existing network. When networks get large, new problems arise. Cities can have traffic jams, a shortage of telephone numbers, and it is easy to get lost. Designs that continue to work well when the network gets large are said to be **scalable**.

Since there are many computers on the network, every layer needs a mechanism for identifying the senders and receivers that are involved in a particular message. This mechanism is called **addressing** or **naming**.

A third design issue is resource allocation. Networks provide a service to hosts from their underlying resources, such as the capacity of transmission lines. To do this well, they need mechanisms that divide their resources so that one host does not interfere with another too much. Many designs share network bandwidth dynamically, according to the short term needs of hosts, rather than by giving each host a fixed fraction of the bandwidth that it may or may not use. This design is called **statistical multiplexing**, meaning sharing based on the statistics of demand.

An allocation problem that occurs at every level is how to keep a fast sender from swamping a slow receiver with data. For this Feedback from the receiver to the sender is often used. It is

called **Flow control**. Sometimes the problem is that the network is oversubscribed because too many computers want to send too much traffic, and the network cannot deliver it all. This overloading of the network is called **congestion**. One strategy is for each computer to reduce its demand when it experiences congestion.

The last major **design issue is to secure the network by defending it against different kinds of threats**. One of the threats we have mentioned previously is that of eavesdropping on communications. Mechanisms that provide confidentiality defend against this threat, and they are used in multiple layers. Mechanisms for authentication prevent someone from impersonating someone else.

1.3.3 Connection-Oriented Versus Connectionless Service

Layers can offer two different types of service to the layers above them: **connection-oriented** and **connectionless**.

<u>Connection Oriented Services</u>	<u>Connection Less Service</u>
They are designed on the basis of the Telephone system i.e; , to use a connection-oriented network service, the service user first establishes a connection, uses the connection, and then releases the connection	Based on the Postal system
In this type of service, Prior connection needs to be established	No prior connection is needed
These services ensure reliable transfer of data	Reliability is not guaranteed
No possibility of congestion	There are chances of occurrence of congestion using these services.
In this authentication is required before transmitting the data packets to the receiver	Authentication is not required before transmitting the data packets to the receiver
Suitable for long and steady transmissions	Suitable for bursty transmissions
Connection is established through process of signalling	No such signalling concept exists
Data packets travel towards their destination node in a sequential manner	Data packets travel towards their destination node in a random manner
Retransmission of lost data bits is possible	Retransmission is Not possible

1.3.4 Service Primitives

A service is formally specified by a set of **primitives** (operations) available to user processes to access the service. These primitives tell the service to perform some action or report on an action taken by a peer entity. The set of primitives available depends on the nature of the service being provided. The primitives for connection-oriented service are different from those of connectionless service.

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
ACCEPT	Accept an incoming connection from a peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Figure 1-17. Six service primitives that provide a simple connection-oriented service.

These primitives might be used for a request-reply interaction in a client-server environment. It is illustrated as follows. First, the server executes LISTEN to indicate that it is prepared to accept incoming connections. After executing the primitive, the server process is blocked until a request for connection appears.

Next, the client process executes CONNECT to establish a connection with the server. The CONNECT call needs to specify who to connect to, so it might have a parameter giving the server's address. The operating system then typically sends a packet to the peer asking it to connect, as shown by (1) in Fig. 1-18. The client process is suspended until there is a response.

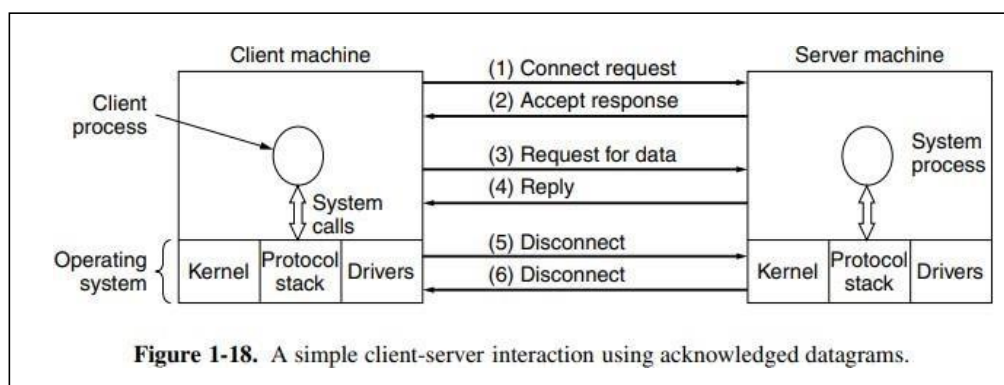


Figure 1-18. A simple client-server interaction using acknowledged datagrams.

When the packet arrives at the server, the operating system sees that the packet is requesting a connection. It checks to see if there is a listener, and if so it unblocks the listener. The server process can then establish the connection with the ACCEPT call. This sends a response (2)

back to the client process to accept the connection. The arrival of this response then releases the client. At this point the client and server are both running and they have a connection established.

The next step is for the server to execute `RECEIVE` to prepare to accept the first request. The `RECEIVE` call blocks the server.

Then the client executes `SEND` to transmit its request (3) followed by the execution of `RECEIVE` to get the reply. The arrival of the request packet at the server machine unblocks the server so it can handle the request. After it has done the work, the server uses `SEND` to return the answer to the client (4). The arrival of this packet unblocks the client, which can now inspect the answer. If the client has additional requests, it can make them now.

When the client is done, it executes `DISCONNECT` to terminate the connection (5). Usually, an initial `DISCONNECT` is a blocking call, suspending the client and sending a packet to the server saying that the connection is no longer needed. When the server gets the packet, it also issues a `DISCONNECT` of its own, acknowledging the client and releasing the connection (6). When the server's packet gets back to the client machine, the client process is released and the connection is broken. In a nutshell, this is how connection-oriented communication works.

1.3.5 The Relationship of Services to Protocols

A **service** is a set of primitives (operations) that a layer provides to the layer above it. The service defines what operations the layer is prepared to perform on behalf of its users, but it says nothing at all about how these operations are implemented. A service relates to an interface between two layers, with the lower layer being the service provider and the upper layer being the service user.

A **protocol**, in contrast, is a set of rules governing the format and meaning of the packets, or messages that are exchanged by the peer entities within a layer. Entities use protocols to implement their service definitions. They are free to change their protocols at will, provided they do not change the service visible to their users.

1.4 REFERENCE MODELS

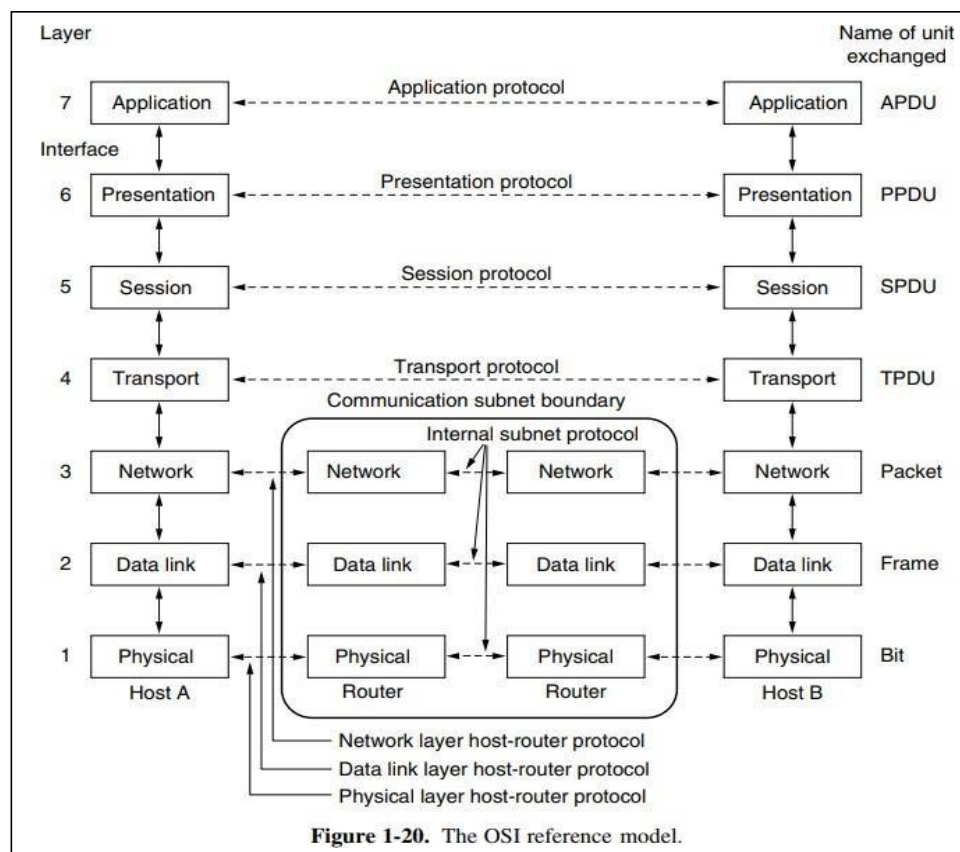
We will discuss two important network architectures: the OSI reference model and the TCP/IP reference model.

1.4.1 The OSI Reference Model

The OSI model (minus the physical medium) is shown in Fig. 1-20. This model is based on a proposal developed by the International Standards Organization (ISO). The model is called the **ISO OSI (Open Systems Interconnection) Reference Model** because it deals with connecting open systems—that is, systems that are open for communication with other systems. We will just call it the OSI model for short.

The OSI model has seven layers. The principles that were applied to arrive at the seven layers can be briefly summarized as follows:

1. A layer should be created where a different abstraction is needed.
2. Each layer should perform a well-defined function.
3. The function of each layer should be chosen with an eye toward defining internationally standardized protocols.



4. The layer boundaries should be chosen to minimize the information flow across the interfaces.

5. The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy

The Physical Layer

The **physical layer is concerned with transmitting raw bits over a communication channel.**

The design issues have to do with making sure that when one side sends a 1 bit it is received by the other side as a 1 bit, not as a 0 bit. Typical questions here are what electrical signals should be used to represent a 1 and a 0, how many nanoseconds a bit lasts, whether transmission may proceed simultaneously in both directions, how the initial connection is established, how it is torn down when both sides are finished, how many pins the network connector has, and what each pin is used for. These design issues largely deal with mechanical, electrical, and timing interfaces, as well as the physical transmission medium, which lies below the physical layer.

The Data Link Layer

The main task of the data link layer is to **transform a raw transmission facility into a line that appears free of undetected transmission errors.** It does so by masking the real errors so the network layer does not see them. It accomplishes this task by having the sender break up the input data into data frames (typically a few hundred or a few thousand bytes) and transmit the frames sequentially. If the service is reliable, the receiver confirms correct receipt of each frame by sending back an acknowledgement frame.

The Network Layer

The network layer controls the operation of the subnet. A key design issue is determining how packets are routed from source to destination. Routes can be based on static tables that are “wired into” the network and rarely changed, or more often they can be updated automatically to avoid failed components. They can also be determined at the start of each conversation.

If too many packets are present in the subnet at the same time, they will get in one another’s way, forming bottlenecks. Handling congestion is also a responsibility of the network layer, in conjunction with higher layers that adapt the load they place on the network. More generally, the quality of service provided (delay, transit time, jitter, etc.) is also a network layer issue. When a packet has to travel from one network to another to get to its destination, many problems can arise. The addressing used by the second network may be different from that used

by the first one. The second one may not accept the packet at all because it is too large. The protocols may differ, and so on. It is up to the network layer to overcome all these problems to allow heterogeneous networks to be interconnected.

The Transport Layer

The basic function of the transport layer is to **accept data from above it, split it up into smaller units, if need be, pass these to the network layer, and ensure that the pieces all arrive correctly at the other end.**

The transport layer also determines what type of service to provide to the session layer, and, ultimately, to the users of the network. The most popular type of transport connection is an error-free point-to-point channel that delivers messages or bytes in the order in which they were sent. The type of service is determined when the connection is established.

The transport layer is a true end-to-end layer; it carries data all the way from the source to the destination. In other words, a program on the source machine carries on a conversation with a similar program on the destination machine, using the message headers and control messages.

The Session Layer

The session layer allows users on different machines to establish sessions between them. Sessions offer various services, including dialog control (keeping track of whose turn it is to transmit), token management (preventing two parties from attempting the same critical operation simultaneously), and synchronization (checkpointing long transmissions to allow them to pick up from where they left off in the event of a crash and subsequent recovery).

The Presentation Layer

The presentation layer is concerned with the **syntax and semantics of the information transmitted.** In order to make it possible for computers with different internal data representations to communicate, the data structures to be exchanged can be defined in an abstract way, along with a standard encoding to be used “on the wire.” The presentation layer manages these abstract data structures and allows higher-level data structures (e.g., banking records) to be defined and exchanged.

The Application Layer

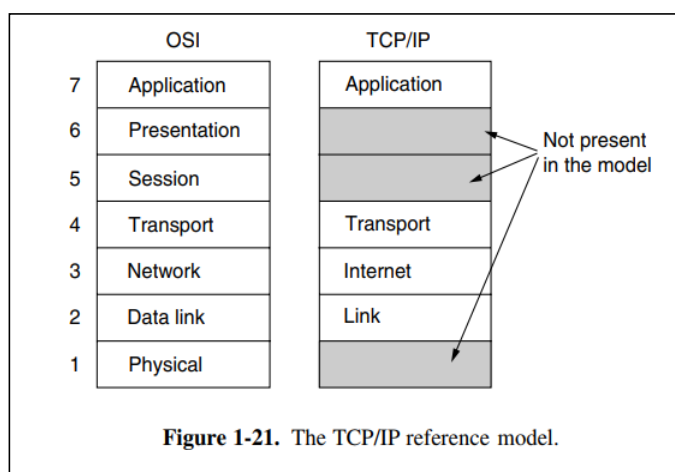
The application layer contains a variety of protocols that are commonly needed by users. One widely used application protocol is HTTP (HyperText Transfer Protocol), which is the basis

for the World Wide Web. When a browser wants a Web page, it sends the name of the page it wants to the server hosting the page using HTTP. The server then sends the page back.

1.4.2 The TCP/IP Reference Model

The Link Layer

Major goal was that the network be able to survive loss of subnet hardware, without existing conversations being broken off. In other words, the DoD (U.S. Department of Defence) wanted connections to remain intact as long as the source and destination machines were functioning, even if some of the machines or transmission lines in between were suddenly put out of operation.



All these requirements led to the choice of a packet-switching network based on a connectionless layer that runs across different networks. The lowest layer in the model, the link layer describes what links such as serial lines and classic Ethernet must do to meet the needs of this connectionless internet layer. It is not really a layer at all, in the normal sense of the term, but rather an interface between hosts and transmission links.

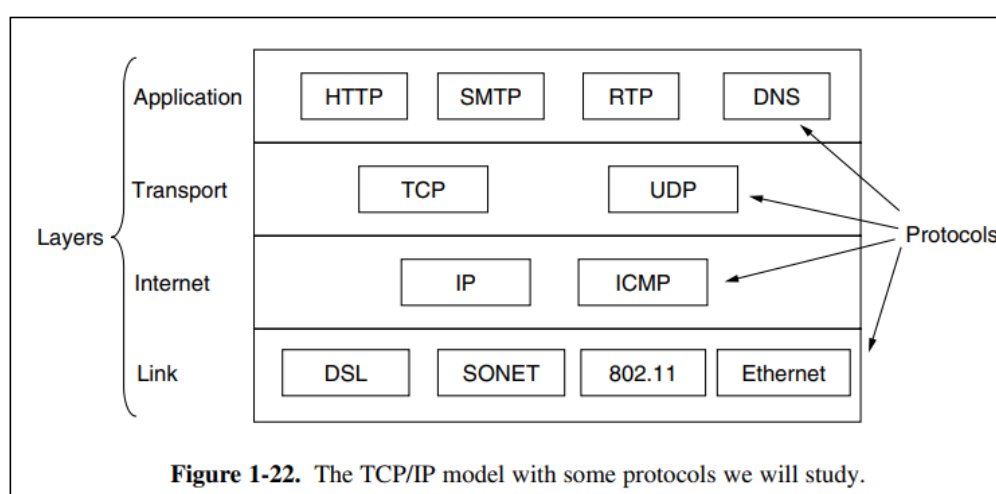
The Internet Layer

The internet layer is the linchpin that holds the whole architecture together. It is shown in Fig. 1-21 as corresponding roughly to the OSI network layer. Its job is to permit hosts to inject packets into any network and have them travel independently to the destination (potentially on a different network). They may even arrive in a completely different order than they were sent, in which case it is the job of higher layers to rearrange them, if in-order delivery is desired.

The internet layer defines an official packet format and protocol called IP (Internet Protocol), plus a companion protocol called ICMP (Internet Control Message Protocol) that helps it function. The job of the internet layer is to deliver IP packets where they are supposed to go.

The Transport Layer

The layer above the internet layer in the TCP/IP model is now usually called the transport layer. It is designed to allow peer entities on the source and destination hosts to carry on a conversation, just as in the OSI transport layer. Two end-to-end transport protocols have been defined here. The first one, **TCP (Transmission Control Protocol)**, is a reliable connection-oriented protocol that allows a byte stream originating on one machine to be delivered without error on any other machine in the internet. It segments the incoming byte stream into discrete messages and passes each one on to the internet layer. At the destination, the receiving TCP process reassembles the received messages into the output stream. TCP also handles flow control to make sure a fast sender cannot swamp a slow receiver with more messages than it can handle.



The second protocol in this layer, **UDP (User Datagram Protocol)**, is an unreliable, connectionless protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own. It is also widely used for one-shot, client-server-type request-reply queries and applications in which prompt delivery is more important than accurate delivery, such as transmitting speech or video. The relation of IP, TCP, and UDP is shown in Fig. 1-22

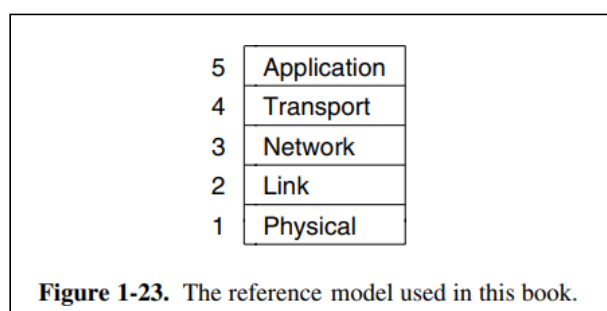
The Application Layer

The TCP/IP model does not have session or presentation layers. On top of the transport layer is the application layer. It contains all the higher-level protocols. The early ones included virtual terminal (TELNET), file transfer (FTP), and electronic mail (SMTP). Many other protocols have been added to these over the years. Some important ones are shown in Fig. 1-22, include the Domain Name System (DNS), for mapping host names onto their network

addresses, HTTP, the protocol for fetching pages on the World Wide Web, and RTP, the protocol for delivering real-time media such as voice or movies.

1.4.3 The Hybrid model/Reference Model

This model has five layers, running from the physical layer up through the link, network and transport layers to the application layer. The physical layer specifies how to transmit bits across different kinds of media as electrical (or other analog) signals. The link layer is concerned with how to send finite-length messages between directly connected computers with specified levels of reliability.



The network layer deals with how to combine multiple links into networks, and networks of networks, into internetworks so that we can send packets between distant computers. This includes the task of finding the path along which to send the packets.

The transport layer strengthens the delivery guarantees of the Network layer, usually with increased reliability, and provide delivery abstractions, such as a reliable byte stream, that match the needs of different applications.

Finally, the application layer contains programs that make use of the network.

PHYSICAL LAYER

Physical layer is the lowest layer in the protocol model. It defines the electrical, timing and other interfaces by which bits are sent as signals over channels. The physical layer is the foundation on which the network is built.

2.1 Guided Transmission Media

The purpose of the physical layer is to transport bits from one machine to another. Various physical media can be used for the actual transmission. Media are roughly grouped into guided

media, such as copper wire and fiber optics, and unguided media, such as terrestrial wireless, satellite, and lasers through the air.

2.1.1 Magnetic Media

One of the most common ways to transport data from one computer to another is to write them onto magnetic tape or removable media (e.g., recordable DVDs), physically transport the tape or disks to the destination machine, and read them back in again. This method is cost-effective, especially for applications in which high bandwidth or cost per bit transported is the key factor.

2.1.2 Twisted Pairs

One of the oldest and still most common transmission media is twisted pair. A twisted pair consists of two insulated copper wires, typically about 1 mm thick. The wires are twisted together in a helical form, just like a DNA molecule. Twisting is done because two parallel wires constitute a fine antenna. When the wires are twisted, the waves from different twists cancel out, so the wire radiates less effectively. A signal is usually carried as the difference in voltage between the two wires in the pair. This provides better immunity to external noise because the noise tends to affect both wires the same, leaving the differential unchanged.

The most common application of the twisted pair is the telephone system. Nearly all telephones are connected to the telephone company (telco) office by a twisted pair. Both telephone calls and ADSL Internet access run over these lines. Twisted pairs can run several kilometers without amplification, but for longer distances the signal becomes too attenuated and repeaters are needed. When many twisted pairs run in parallel for a substantial distance, such as all the wires coming from an apartment building to the telephone company office, they are bundled together and encased in a protective sheath. The pairs in these bundles would interfere with one another if it were not for the twisting.

Twisted pairs can be used for transmitting either analog or digital information. The bandwidth depends on the thickness of the wire and the distance traveled, but several megabits/sec can be achieved for a few kilometers in many cases. Due to their adequate performance and low cost, twisted pairs are widely used

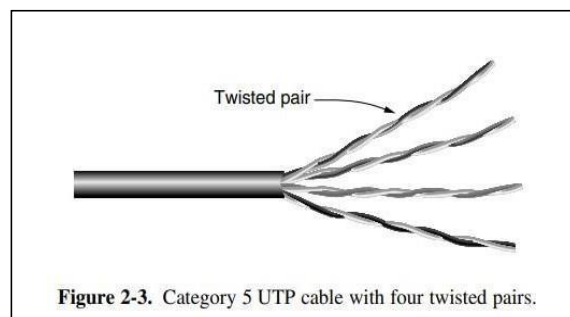


Figure 2-3. Category 5 UTP cable with four twisted pairs.

Twisted-pair cabling comes in several varieties. A **category 5 (Cat 5)** twisted pair consists of two insulated wires gently twisted together. Four such pairs are typically grouped in a plastic sheath to protect the wires and keep them together. This arrangement is shown in Fig. 2-3.

Cat 5 replaced earlier **Category 3** cables with a similar cable that uses the same connector, but has more twists per meter. More twists result in less crosstalk and a better-quality signal over longer distances, making the cables more suitable for high-speed computer communication, especially 100-Mbps and 1-Gbps Ethernet LANs.

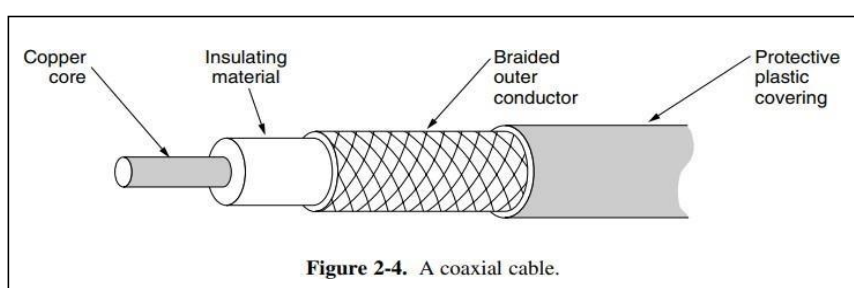
Category 6 or even **Category 7** has more stringent specifications to handle signals with greater bandwidths. Some cables in Category 6 and above are rated for signals of 500 MHz and can support the 10-Gbps links. Category 6 wiring types are referred to as UTP (Unshielded Twisted Pair) as they consist simply of wires and insulators. In contrast to these, Category 7 cables have shielding on the individual twisted pairs, as well as around the entire cable (but inside the plastic protective sheath).

Different LAN standards may use the twisted pairs differently. For example, 100-Mbps Ethernet uses two (out of the four) pairs, one pair for each direction. To reach higher speeds, 1-Gbps Ethernet uses all four pairs in both directions simultaneously;

Links that can be used in both directions at the same time are called **full-duplex links**. In contrast, links that can be used in either direction, but only one way at a time, are called **half-duplex links**. A third category consists of links that allow traffic in only one direction, are called **simplex links**.

2.1.3 Coaxial Cable

Another common transmission medium is the coaxial cable. It has better shielding and greater bandwidth than unshielded twisted pairs, so it can span longer distances at higher speeds. Two kinds of coaxial cable are widely used. One kind, 50-ohm cable, is commonly used when it is intended for digital transmission from the start. The other kind, 75-ohm cable, is commonly used for analog transmission and cable television.

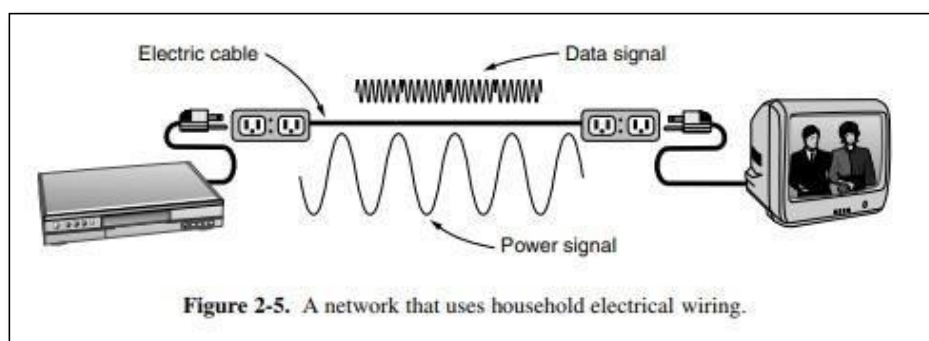


A coaxial cable consists of a stiff copper wire as the core, surrounded by an insulating material. The insulator is encased by a cylindrical conductor, often as a closely woven braided mesh. The outer conductor is covered in a protective plastic sheath.

The coaxial cable has high bandwidth and excellent noise immunity. The bandwidth depends on the cable quality and length. Modern cables have a bandwidth of up to a few GHz. Coax or Coaxial cables is still widely used for cable television and metropolitan area networks.

2.1.4 Power Lines

Power lines deliver electrical power to houses, and electrical wiring within houses distributes the power to electrical outlets. The use of power lines for data communication is an old idea. Power lines have been used by electricity companies for low-rate communication such as remote metering for many years, as well in the home to control devices (e.g., the X10 standard). The convenience of using power lines for networking should be clear. Simply plug a TV and a receiver into the wall, which you must do anyway because they need power, and they can send and receive movies over the electrical wiring. This configuration is shown in Fig. 2-5. There is no other plug or radio. The data signal is superimposed on the low-frequency power signal (on the active or “hot” wire) as both signals use the wiring at the same time.



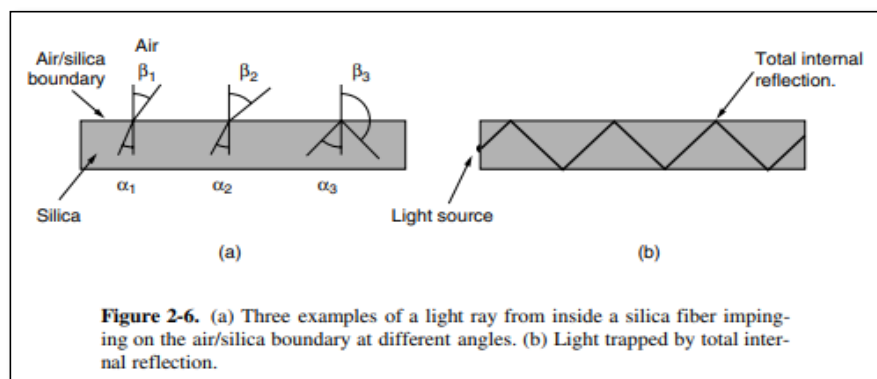
The difficulty with using household electrical wiring for a network is that it was designed to distribute power signals. This task is quite different than distributing data signals, at which household wiring does a horrible job. Electrical signals are sent at 50–60 Hz and the wiring attenuates the much higher frequency (MHz) signals needed for high-rate data communication. The electrical properties of the wiring vary from one house to the next and change as appliances are turned on and off, which causes data signals to bounce around the wiring. Transient currents when appliances switch on and off create electrical noise over a wide range of frequencies. And without the careful twisting of twisted pairs, electrical wiring acts as a fine antenna, picking up external signals and radiating signals of its own.

Despite these difficulties, it is practical to send at least 100 Mbps over typical household electrical wiring by using communication schemes that resist impaired frequencies and bursts of errors.

2.1.5 Fiber Optics

Fiber optics are used for long-haul transmission in network backbones, highspeed LANs (although so far, copper has always managed catch up eventually), and high-speed Internet access such as FttH (Fiber to the Home). An optical transmission system has three key components: the light source, the transmission medium, and the detector. Conventionally, a pulse of light indicates a 1 bit and the absence of light indicates a 0 bit. The transmission medium is an ultra-thin fiber of glass. The detector generates an electrical pulse when light falls on it. By attaching a light source to one end of an optical fiber and a detector to the other, we have a unidirectional data transmission system that accepts an electrical signal, converts and transmits it by light pulses, and then reconverts the output to an electrical signal at the receiving end.

This transmission system would leak light and be useless in practice were it not for an interesting principle of physics. When a light ray passes from one medium to another—for example, from fused silica to air—the ray is refracted (bent) at the silica/air boundary, as shown in Fig. 2-6(a). Here we see a light ray incident on the boundary at an angle α_1 emerging at an angle β_1 . The amount of refraction depends on the properties of the two media (in particular, their indices of refraction). For angles of incidence above a certain critical value, the light is refracted back into the silica; none of it escapes into the air. Thus, a light ray incident at or above the critical angle is trapped inside the fiber, as shown in Fig. 2-6(b), and can propagate for many kilometers with virtually no loss.

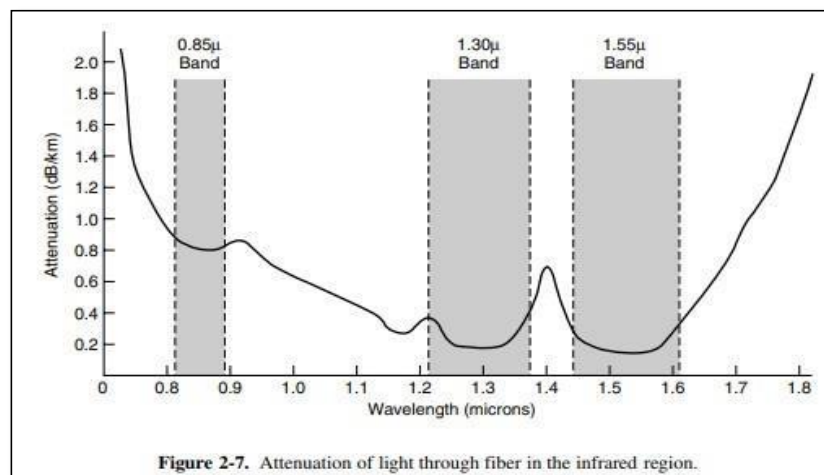


The sketch of Fig. 2-6(b) shows only one trapped ray, but since any light ray incident on the boundary above the critical angle will be reflected internally, many different rays will be bouncing around at different angles. Each ray is said to have a different mode, so a fiber having this property is called a **multimode fiber**.

However, if the fiber's diameter is reduced to a few wavelengths of light the fiber acts like a wave guide and the light can propagate only in a straight line, without bouncing, yielding a single-mode fiber. Currently available single-mode fibers can transmit data at 100 Gbps for 100 km without amplification.

Transmission of Light Through Fiber

Optical fibers are made of glass, which, in turn, is made from sand, an inexpensive raw material available in unlimited amounts. The glass used for modern optical fibers is so transparent. The attenuation of light through glass depends on the wavelength of the light (as well as on some physical properties of the glass). It is defined as the ratio of input to output signal power. For the kind of glass used in fibers, the attenuation is shown in Fig. 2-7 in units of decibels per linear kilometer of fiber.

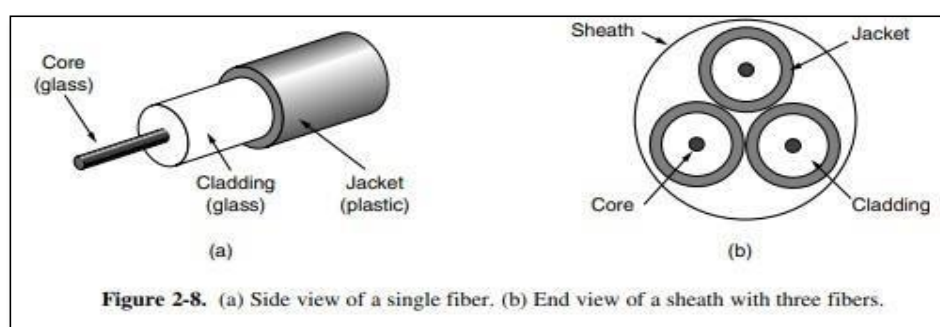


Three wavelength bands are most commonly used at present for optical communication. They are centered at 0.85, 1.30, and 1.55 microns, respectively. All three bands are 25,000 to 30,000 GHz wide. The 0.85-micron band was used first. It has higher attenuation and so is used for shorter distances. The last two bands have good attenuation properties (less than 5% loss per kilometer). The 1.55-micron band is now widely used with erbium-doped amplifiers that work directly in the optical domain.

Light pulses sent down a fiber spread out in length as they propagate. This spreading is called **chromatic dispersion**. The amount of it is wavelength dependent. One way to keep these spread-out pulses from overlapping is to increase the distance between them, but this can be done only by reducing the signaling rate. It is also possible to send pulses for thousands of kilometers without appreciable shape distortion. These pulses are called **solitons**.

Fiber Cables

Fiber optic cables are similar to coax, except without the braid. Figure 2-8(a) shows a single fiber viewed from the side. At the center is the glass core through which the light propagates. In multimode fibers, the core is typically 50 microns in diameter, about the thickness of a human hair. In single-mode fibers, the core is 8 to 10 microns.



The core is surrounded by a glass cladding with a lower index of refraction than the core, to keep all the light in the core. Next comes a thin plastic jacket to protect the cladding. Fibers are typically grouped in bundles, protected by an outer sheath. Figure 2-8(b) shows a sheath with three fibers.

Fibers can be connected in three different ways. First, they can terminate in connectors and be plugged into fiber sockets. Second, they can be spliced mechanically. Mechanical splices just lay the two carefully cut ends next to each other in a special sleeve and clamp them in place. Alignment can be improved by passing light through the junction and then making small adjustments to maximize the signal. Third, two pieces of fiber can be fused (melted) to form a solid connection. For all three kinds of splices, reflections can occur at the point of the splice, and the reflected energy can interfere with the signal.

Two kinds of light sources are typically used to do the signaling. These are LEDs (Light Emitting Diodes) and semiconductor lasers. They have different properties. They can be tuned in wavelength by inserting Fabry-Perot or Mach-Zehnder interferometers between the source

and the fiber. Fabry-Perot interferometers are simple resonant cavities consisting of two parallel mirrors. The light is incident perpendicular to the mirrors. The length of the cavity selects out those wavelengths that fit inside an integral number of times. Mach-Zehnder interferometers separate the light into two beams. The two beams travel slightly different distances. They are recombined at the end and are in phase for only certain wavelengths.

The receiving end of an optical fiber consists of a photodiode, which gives off an electrical pulse when struck by light. Issue lies in Thermal noise, so a pulse of light must carry enough energy to be detected. By making the pulses powerful enough, the error rate can be made arbitrarily small.

Comparison of Fibre Optics and Copper Wire

	Fiber Optics	Copper Wire
1.	It can handle much higher bandwidths than copper	It can not handle higher bandwidths like Fibre optics does
2.	Due to low attenuation, repeaters are needed only about every 50 km on long lines	Whereas repeaters are needed for every 5km in case of copper wire
3.	Fibre has the advantage of not being affected by power surges, electromagnetic interference or power failures and also not affected by corrosive chemicals in the air	Affected by environmental factors unlike Fibre Optics
4.	It is thin and lightweight	Not lighter in weight when compared to Fibre optics

2.2 Wireless Transmission

2.2.1 The Electromagnetic Spectrum

When an antenna of the appropriate size is attached to an electrical circuit, the electromagnetic waves can be broadcast efficiently and received by a receiver some distance away. All wireless communication is based on this principle.

In a vacuum, all electromagnetic waves travel at the same speed, no matter what their frequency. This speed, usually called the speed of light, c , is approximately 3×10^8 m/sec. In copper or fiber the speed slows to about 2/3 of this value and becomes slightly frequency dependent. The fundamental relation between f , λ , and c (in a vacuum) is

$$\lambda f = c$$

The electromagnetic spectrum is shown in Fig. 2-10. The radio, microwave, infrared, and visible light portions of the spectrum can all be used for transmitting information by modulating the amplitude, frequency, or phase of the waves. Ultraviolet light, X-rays, and gamma rays would be even better, due to their higher frequencies, but they are hard to produce and modulate, do not propagate well through buildings, and are dangerous to living things. In the figure shown below, The terms LF, MF, and HF refer to Low, Medium, and High Frequency, respectively. Also the higher bands are named as Very, Ultra, Super, Extremely, and Tremendously High Frequency bands.

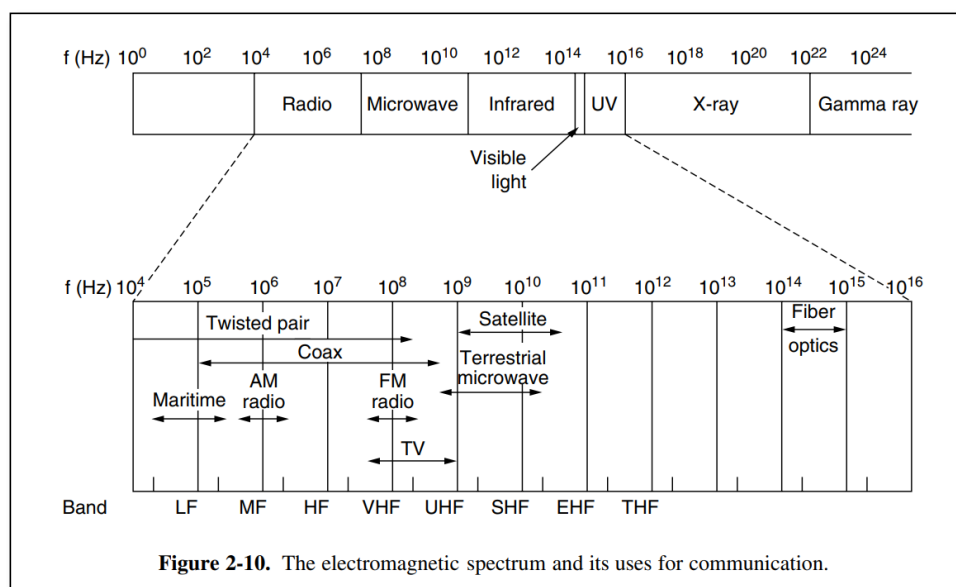


Figure 2-10. The electromagnetic spectrum and its uses for communication.

Most transmissions use a relatively narrow frequency band (i.e., $\Delta f/f \ll 1$). They concentrate their signals in this narrow band to use the spectrum efficiently and obtain reasonable data rates by transmitting with enough power. However, in some cases, a wider band is used, with three variations. In **frequency hopping spread spectrum**, the transmitter hops from frequency to frequency hundreds of times per second. It is popular for military communication because it makes transmissions hard to detect and next to impossible to jam.

A second form of spread spectrum, **direct sequence spread spectrum**, uses a code sequence to spread the data signal over a wider frequency band. It is widely used commercially as a spectrally efficient way to let multiple signals share the same frequency band. These signals can be given different codes, a method called CDMA (Code Division Multiple Access).

A third method of communication with a wider band is **UWB (Ultra WideBand)** communication. UWB sends a series of rapid pulses, varying their positions to communicate

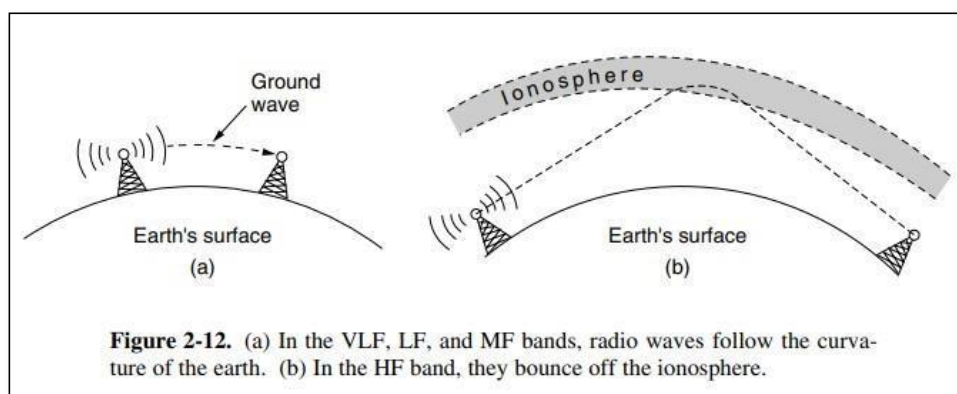
information. The rapid transitions lead to a signal that is spread thinly over a very wide frequency band

2.2.2 Radio Transmission

Radio frequency (RF) waves are easy to generate, can travel long distances, and can penetrate buildings easily, so they are widely used for communication, both indoors and outdoors. Radio waves also are omnidirectional, meaning that they travel in all directions from the source, so the transmitter and receiver do not have to be carefully aligned physically.

The properties of radio waves are frequency dependent. At low frequencies, radio waves pass through obstacles well, but the power falls off sharply with distance from the source. This attenuation is called path loss. At high frequencies, radio waves tend to travel in straight lines and bounce off obstacles.

In the VLF, LF, and MF bands, radio waves follow the ground, as illustrated in Fig. 2-12(a). These waves can be detected for perhaps 1000 km at the lower frequencies, less at the higher ones. Radio waves in these bands pass through buildings easily, which is why portable radios work indoors. The main problem with using these bands for data communication is their low bandwidth.



In the HF and VHF bands, the ground waves tend to be absorbed by the earth. However, the waves that reach the ionosphere, a layer of charged particles circling the earth at a height of 100 to 500 km, are refracted by it and sent back to earth, as shown in Fig. 2-12(b). Under certain atmospheric conditions, the signals can bounce several times.

2.2.3 Microwave Transmission

Above 100 MHz, the waves travel in nearly straight lines and can therefore be narrowly focused. Concentrating all the energy into a small beam by means of a parabolic antenna (like

the familiar satellite TV dish) gives a much higher signal-to-noise ratio, but the transmitting and receiving antennas must be accurately aligned with each other. In addition, this directionality allows multiple transmitters lined up in a row to communicate with multiple receivers in a row without interference, provided some minimum spacing rules are observed.

Unlike radio waves at lower frequencies, microwaves do not pass through buildings well. In addition, even though the beam may be well focused at the transmitter, there is still some divergence in space. Some waves may be refracted off low-lying atmospheric layers and may take slightly longer to arrive than the direct waves. The delayed waves may arrive out of phase with the direct wave and thus cancel the signal. This effect is called multipath fading and is often a serious problem. It is weather and frequency dependent.

Microwave communication is so widely used for long-distance telephone communication, mobile phones, television distribution, and other purposes that a severe shortage of spectrum has developed. It has several key advantages over fiber. There is no need to lay down cables. It is also relatively inexpensive.

2.3.4 Infrared Transmission

Unguided infrared waves are widely used for short-range communication. The remote controls used for televisions, VCRs, and stereos all use infrared communication. They are relatively directional, cheap, and easy to build but have a major drawback: they do not pass through solid objects.

Security of infrared systems against eavesdropping is better than that of radio systems. Therefore, no government license is needed to operate an infrared system, in contrast to radio systems, which must be licensed outside the ISM bands.

2.3.5 Light Transmission

Unguided optical signaling or free-space optics has been in use for centuries. A more modern application is to connect the LANs in two buildings via lasers mounted on their rooftops. Optical signaling using lasers is inherently unidirectional, so each end needs its own laser and its own photodetector. This scheme offers very high bandwidth at very low cost and is relatively secure because it is difficult to tap a narrow laser beam. It is also relatively easy to install and, unlike microwave transmission, does not require an FCC license.

Wind and temperature changes can distort the beam and laser beams also cannot penetrate rain or thick fog, although they normally work well on sunny days.

